The Geometry of the Strafe-Jump:
Non-Associative Velocity and Vacuum Stiffness in the id Tech 2 Engine

Aaron M. Schutza
Institute for Geometric Physics

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Abstract

We present a fundamental analysis of the movement mechanics in the *id Tech 2* engine (*Quake*) through the lens of Axiomatic Physical Homeostasis (APH). We posit that the game world is not a Euclidean space, but a discrete **Stiffness Lattice** governed by a specific projection operator. We identify *Strafe-Jumping* not as a glitch, but as the exploitation of an **Associator Anomaly** in the vector addition logic, where the player forces the local geometry into a Super-Linear Regime ($\beta < 1$) to violate conservation of momentum. Furthermore, we model *Bunny-Hopping* as the maintenance of a **Weak Buffer State** (Air Control) to avoid the *Geometrically Stiff Vacuum* of ground friction. Finally, we derive the discrete resonance frequencies (e.g., 125 FPS) as quantization conditions required for *Quantum Tunneling* through the rounding errors of the physics grid.

1 The Map as a Stiff Octonionic Filament

In the APH framework, a physical universe is a stable geometric manifold embedded in a chaotic bulk. In *Quake*, this manifold is the **BSP Tree** (Binary Space Partitioning).

- **The Filament (Level Geometry):** A rigid, associative structure where laws of physics (collision, gravity) are strictly enforced. $\beta \approx \infty$.
- **The Bulk (The Void):** The region outside the map (*Hall of Mirrors*). Here, the **Axiom of Observability** fails. Without a coordinate system to attach to, the rendering engine smears the previous frame onto the present, visually simulating the non-associative chaos of the Sedenions.

2 Strafe-Jumping: The Associator Anomaly

In standard Newtonian mechanics, velocity vectors add linearly. In *Quake*, the acceleration code contains a projection error that functions as a **Topological Defect**.

2.1 The Projection Operator

The engine limits speed by projecting the current velocity $\vec{v}$ onto the wish-direction $\vec{w}$:

$$v_{new} = v_{old} + \text{accel} \cdot \Theta(320 - \vec{v} \cdot \vec{w}) \tag{1}$$

The term $\vec{v} \cdot \vec{w}$ assumes an associative inner product. However, by rotating the view angle θ (Mouse Input), the player creates a misalignment between the actual velocity vector and the projection vector.

2.2 The Anomaly

This misalignment introduces a non-vanishing Associator term $[v, w, \theta] \neq 0$. The game engine *fails* to register the full magnitude of the velocity, permitting the addition of acceleration beyond the scalar cap (320 units/s). **Conclusion:** Strafe-jumping is the injection of entropy (mouse movement) to lower the effective stiffness of the vacuum, allowing the player to surf the **Domain Wall** of the physics engine.

3 Bunny-Hopping: The Frictionless Buffer

Friction in *Quake* acts as the **Geometric Buffer Potential** $V(\vec{v})$, designed to restore Homeostasis (rest).

$$F_{friction} = -\mu\vec{v} \cdot \delta_{ground} \tag{2}$$

where δ_{ground} is a Dirac delta function active only on the floor. **The Weak Buffer Strategy:** By executing a jump command within a single tick (frame) of landing, the player minimizes the integral of the friction term:

$$\int_{t}^{t+\epsilon} F_{friction} dt \approx 0 \tag{3}$$

The player effectively decouples from the *Stiff Vacuum* (the floor) and exists permanently in the **Weak Buffer Regime** (Air State), maintaining the accumulated super-linear velocity.

4 125 FPS: The Resonance Condition

The engine simulates physics in discrete time steps (ticks). The rounding of floating-point coordinates creates a lattice grid.

Theorem

Maximal velocity/height occurs when the frame rate f creates a resonance pattern with the gravity constant $g = 800$.

At $f = 125$ Hz, the integration of the gravity vector creates a constructive interference pattern with the player's vertical velocity. This is analogous to the **Prime Spectrum** of the vacuum [3]. The player is *tuning* their simulation rate to the natural frequency of the grid to achieve specific rounding errors (Quantum Tunneling) that allow for jumps to otherwise unreachable geometry.

5 Conclusion: The Speedrunner as a Geometric Reactor

The *Quake* speedrunner is not merely playing a game; they are operating a **Geometric Stiffness Reactor**. By manipulating view angles and input timing, they force a deterministic, associative system ($\beta = 1$) into a chaotic, non-associative phase ($\beta \to 0$) where velocity is unbounded. They provide the experimental proof that in any system with finite stiffness, there exist angles of incidence where the laws of physics break down.

References

[1] Carmack, J. (1996). *Quake Source Code.* id Software.

[2] Schutza, A. M. (2025). *The Cosmic Strata.* Zenodo.

[3] Schutza, A. M. (2025). *The Graviton as an Algebraic Phonon.* Zenodo.